

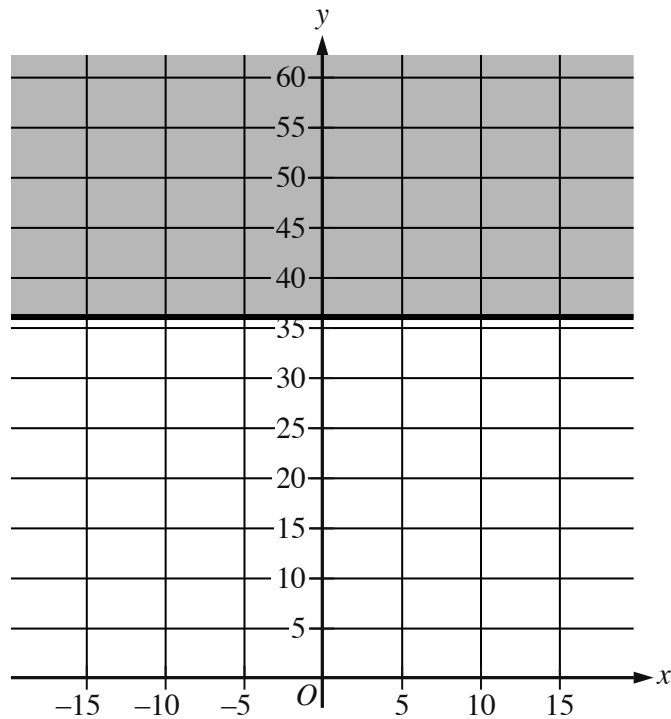
☒ EASY

☐ ALGEBRA

Mixed

10 questions · ~15 min

02



- The boundary of the inequality is a solid line.
 - The line is horizontal.
 - The line passes through the following points:
 - (negative 15 comma 36)
 - (0 comma 36)
 - (15 comma 36)
- The area above the boundary is shaded.

The shaded region shown in the graph represents all the solutions to which inequality?

- A. $x \leq 36$
- B. $x \geq 36$
- C. $y \leq 36$

D. $y \geq 36$

Q2

ALGEBRA

A librarian has 43 books to distribute to a group of children. If he gives each child 2 books, he will have 7 books left over. How many children are in the group?

- A. 15
- B. 18
- C. 25
- D. 29

Q3

ALGEBRA

Hana deposited a fixed amount into her bank account each month. The function $f(t) = 100 + 25t$ gives the amount, in dollars, in Hana's bank account after t monthly deposits. What is the best interpretation of 25 in this context?

- A. With each monthly deposit, the amount in Hana's bank account increased by \$ 25.
- B. Before Hana made any monthly deposits, the amount in her bank account was \$ 25.
- C. After 1 monthly deposit, the amount in Hana's bank account was \$ 25.
- D. Hana made a total of 25 monthly deposits.

Q4

ALGEBRA

If $4x = 3$, what is the value of $24x$?

- A. $\frac{9}{2}$
- B. 6
- C. 18
- D. 72

Q5

PROBLEM-SOLVING AND DATA ANALYSIS

The weight of an object on Venus is approximately $\frac{9}{10}$ of its weight on Earth. The weight of an object on Jupiter is approximately $\frac{23}{10}$ of its weight on Earth. If an object weighs 100 pounds on Earth, approximately how many more pounds does it weigh on Jupiter than it weighs on Venus?

- A. 90
- B. 111
- C. 140
- D. 230

Q6

PROBLEM-SOLVING AND DATA ANALYSIS

6, 6, 8, 8, 8, 10, 21

Which of the following lists represents a data set that has the same median as the data set shown?

- A. 4, 6, 6, 6, 8, 8
- B. 6, 6, 8, 8, 10, 10
- C. 6, 8, 10, 10, 10, 12
- D. 8, 8, 10, 10, 21, 21

Q7

PROBLEM-SOLVING AND DATA ANALYSIS

How many yards are equivalent to 612 inches? 1 yard = 36 inches

- A. 0.059
- B. 17
- C. 576
- D. 22,032

Q8

ADVANCED MATH

Which of the following is equivalent to $3(x+5)-6$?

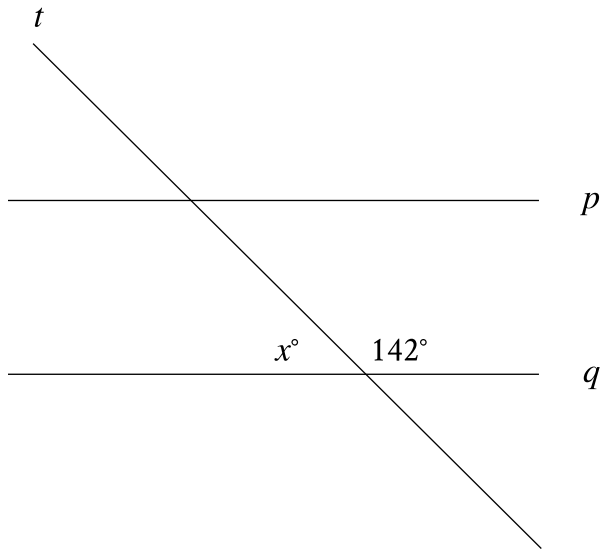
- A. $3x-3$
- B. $3x-1$
- C. $3x+9$
- D. $15x-6$

Q9

ADVANCED MATH

The function g is defined by $gx = \sqrt{8x+1}$. What is the value of $g3$?

- A. $\frac{5}{8}$
- B. $\frac{25}{8}$
- C. 5
- D. 25



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled t , p , and q .
- Line t intersects both line p and line q .
- At the intersection of line t and line q , 2 angles are labeled clockwise from top left as follows:
 - Top left: x°
 - Top right: 142°
- A note indicates the figure is not drawn to scale.

In the figure, line p is parallel to line q , and line t intersects both lines. What is the value of $x + 142$?

- A. 52
- B. 90
- C. 142

D. 180